initAndTerm.c 1.26 KiB

```c
#include <stdio.h>
#include <stdlib.h>
#include "commonType.h"
#include "initAndTerm.h"

int createBFSGraph(graphType *g, int totalCnt) {
    int v;

    if (g->n != 0) {
        releaseNearListHead(g, g->n);
        releaseBFSGraph(g);
    }
    g->n = 0;
    g->nearListHead = (graphNode**)malloc(sizeof(graphNode*)*totalCnt);
    g->visited = (int*)malloc(sizeof(int)*totalCnt);
    for (v = 0; v < totalCnt; v++) {
        g->nearListHead[v] = NULL;
        g->visited[v] = FALSE;
    }
    return 0;
}
int releaseBFSGraph(graphType *g) {
    free(g->nearListHead);
    free(g->visited);
    return 0;
}
int releaseNearListHead(graphType *g, int totalCnt){
    int v;
    graphNode *curP=NULL, *nextP=NULL;
    for(v = 0;v < totalCnt; v++){
        nextP = g->nearListHead[v];
        while(nextP){
            curP = nextP;
            nextP = curP->link;
            free(curP);
        }
    }
}
int createVertexList(int **vList, int totalCnt) {
    *vList = (int*)malloc(sizeof(int)*totalCnt);
    return 0;
}
int releaseVertexList(int **vList) {
    free(*vList);
    return 0;
}
LQType* createLQ(){
    LQType *LQ;
    LQ = (LQType *)malloc(sizeof(LQType));
    LQ->front = NULL;
    LQ->rear = NULL;
    return LQ;
}
int releaseLQ(LQType **LQ){
    free(*LQ);
    return 0;
}

```